MARK User Manual

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Windows x64 English Edition

MARK v1.0.0

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1 Software overview

MARK (Metadata-Guided Automation of Resource Workflows) searches, filters and downloads GNSS observation data and precise products. It queries local file records for specified UTC dates, supports station and product selection by interval, equipment, constellations, date coverage and spatial conditions, then adds the required data to download plans. MARK prepares data for precise positioning, orbit determination, deformation monitoring, and ionospheric and tropospheric research.

This manual covers the MARK v1.0.0 Windows x64 release and its DuckDB knowledge bases, including installation, observation filtering, spatial station selection, product presets, downloading and file management. The overall data cutoff is August 1, 2026, inclusive, but some sources end earlier or contain gaps. Screenshots illustrate operation; actual counts and equipment records depend on dates, sources and criteria.

MARK has two workspaces. Observations searches RINEX observation files by station and date. Precise Products searches orbit, clock, bias, ionospheric map and other auxiliary files. Each workspace maintains its own destination, plan and task states. Users can continue querying and adding tasks during transfers.

MARK prepares processing inputs and does not solve PPP, RTK or ionospheric parameters. Before using downloaded files in other software, check formats, constellation scope, observation codes and product compatibility.

1.1 Main functions

(1) Search several global and regional sources for indexed historical observation files while retaining station source labels.

(2) Filter stations by RINEX version, interval, dated receiver and antenna records, constellations, observation codes and date coverage.

(3) Filter research dates by Kp, Dst and F10.7, and narrow candidates by country / region, coordinate bounds or a map rectangle.

(4) Select an even spread or random subset with a target count, optional distance and daily availability constraints, and daily reports.

(5) Search precise products by category, producing institution, level and coverage; apply built-in presets or save custom product rules.

(6) Manage separate observation and product download plans with pause, resume, cancel, existing-file checks and alternative sources.

(7) Save files by year and day of year or in a flat folder. Optionally extract observation files and convert Compact RINEX to ordinary RINEX.

1.2 Local queries and online downloads

Knowledge bases hold file availability, filenames, equipment, product attributes and download routes, normally without the original GNSS file contents. Queries use the local records without repeatedly visiting remote directories. Downloading subsequently accesses provider servers. A successful query and a successful transfer are distinct stages.

Local records are release snapshots. Remote files can move, disappear or acquire different access requirements. A missing local record can also mean an unscanned date. Interpret available, missing, unknown and incomplete-source messages in light of knowledge base coverage.

2 System requirements and installation

2.1 System requirements

Use the complete installer. Compilation is not required. Before installation, check the operating system, runtime and available disk space against Table 2.1.

Table 2.1 System requirements

Item	Requirement
Operating system	Windows 10 or Windows 11, x64.
Interface runtime	Microsoft Edge WebView2 Evergreen Runtime, x64. If the runtime is missing, install it from Microsoft and restart MARK.
Disk space	The current installer estimates 10.7 GB after installation and requires about 15.7 GB free during setup. Downloads, extraction and conversion need additional space. Follow the values displayed by the installer.
Network	Local queries can run offline. Downloading observation files and products requires a network connection and any access permissions required by the provider.
Other components	The package includes database, transfer and observation postprocessing tools. Python and a separate 7-Zip installation are unnecessary. SFTP sources require the Windows OpenSSH client.
Folder permissions	Installation and download folders must be writable. Routine use normally does not require administrator privileges.

Official WebView2 download: https://developer.microsoft.com/en-us/microsoft-edge/webview2/ . Installing Edge alone or copying only MARK.exe does not ensure that all runtime components and resources are present.

2.2 Download the installer from GitHub

The repository is https://github.com/qiliu2025/MARK . Figure 2.1 marks two download routes: the blue Windows download button in the README and Releases in the right sidebar. Both lead to the complete installer.

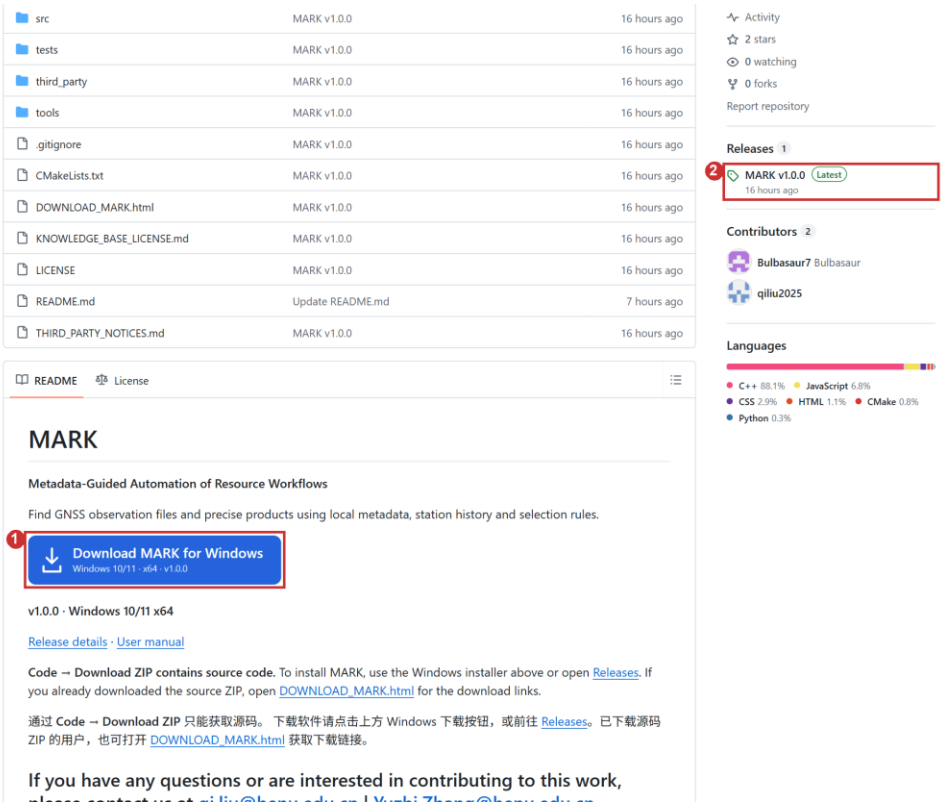


Figure 2.1 Windows download button and Releases on GitHub

(1) Open the repository, scroll to the README and click Download MARK for Windows to obtain the Windows x64 installer.

(2) Alternatively, open Releases, select the version, expand Assets and download MARK-v1.0.0-Windows-x64.exe. Use the actual asset name for newer releases.

(3) Wait for the browser download to finish before running the installer. Code → Download ZIP provides source code, rather than the complete runnable software and knowledge bases.

2.3 Choose the installation folder

The main setup choice is Installation folder. Figure 2.2 shows the actual interface before installation.

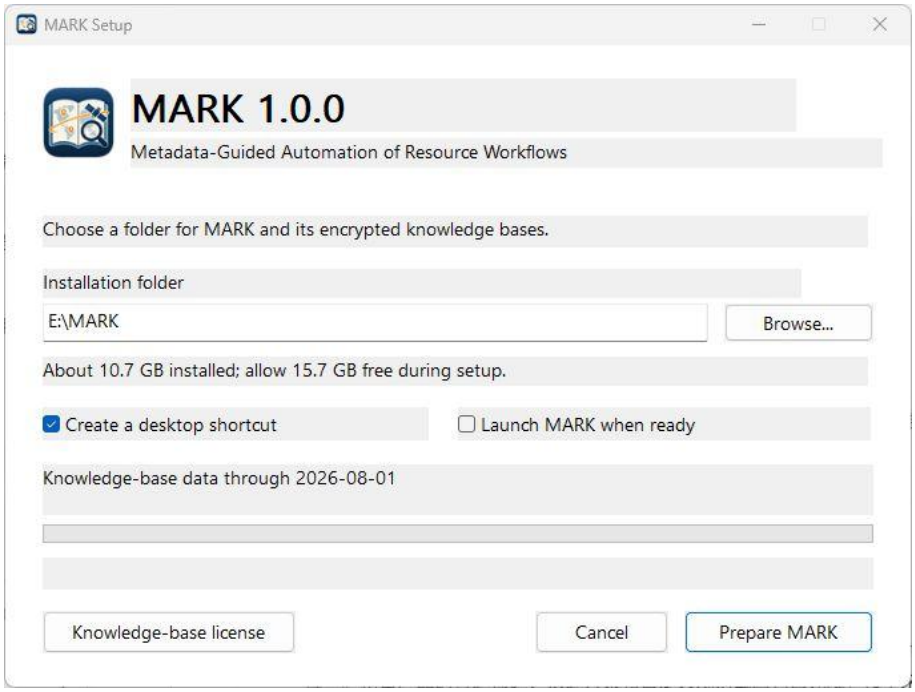


Figure 2.2 Installation folder and Prepare MARK button

(1) Enter the target folder in Installation folder or click Browse... . Prefer a new, empty folder to avoid mixing versions.

(2) Check free space on the target drive. The window displays estimated installed size and space needed during setup, and checks the selected location again.

(3) Optionally select Create a desktop shortcut and Launch MARK when ready. Knowledge-base license opens the knowledge base terms.

(4) Click Prepare MARK and wait for preparation to finish. Do not move the installer or target folder while preparation is running.

(5) Open MARK as instructed, or run software\MARK.exe under the installation folder. If setup fails or space is insufficient, follow the displayed error before retrying.

The installer places the program, knowledge bases and resources together. When moving an installation, move the entire folder and keep software beside knowledge-base. Do not copy only the EXE. A fresh installation starts in English; Chapter 3 explains language settings.

Table 2.2 Main installation contents

Path	Purpose
software\MARK.exe	Desktop application.
software\resources\	Interface, offline maps, reference material and observation postprocessing tools.
software\duckdb.dll and other libraries	Runtime dependencies that must remain with the program.
knowledge-base\	Local knowledge bases and supporting data. Preserve the subfolder structure.
documentation\	User manuals supplied with the package.
source\	Source archive corresponding to the release.

The original software source is provided for noncommercial scientific research. Separate terms apply to the knowledge bases, while third-party components and original data retain their own rights and attribution requirements. See LICENSE, KNOWLEDGE_BASE_LICENSE.md and THIRD_PARTY_NOTICES.md in the package or repository.

3 Settings

3.1 Language and appearance

Click the gear icon in the top bar to open Settings, shown in Figure 3.1. Choose Chinese or English under Language, and Light or Dark under Theme. These options change the interface without changing knowledge base records or filtering results.

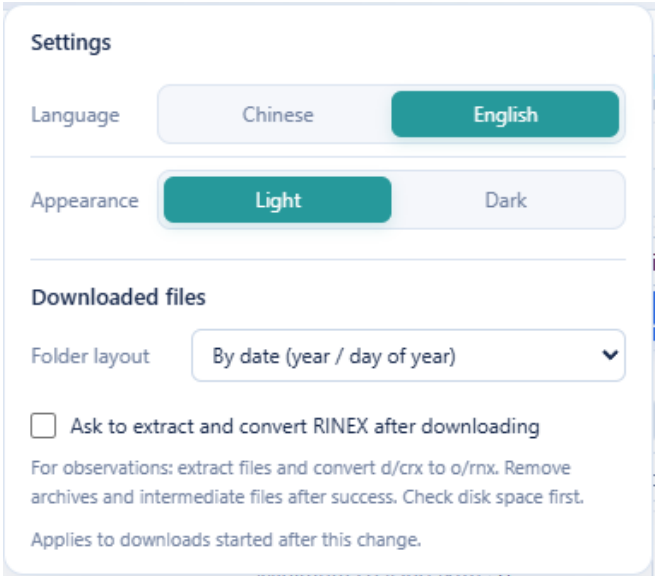


Figure 3.1 Language, theme and download settings

A fresh installation uses English and subsequently remembers its language and theme choices. Click the gear again or outside the panel to close Settings.

3.2 File organization

Save layout controls file organization for tasks started subsequently. The default creates year and day-of-year folders, such as 2024\001 for January 1, 2024. A flat layout writes ordinary files directly to the plan item's destination folder for use by downstream software.

Table 3.1 File organization options

Setting	Observation files	Precise products
Date folders	<observation folder>\YYYY\DOY\<filename>	Usually <product folder>\YYYY\DOY\<category>\<filename>. ANTEX and other specialized products follow their own rules.
Flat destination folder	Ordinary files are written directly to the selected folder.	Ordinary files omit the date and category levels. VMF retains model, grid and solution subfolders to avoid name conflicts.

Changing the setting does not move existing files. The two workspaces have separate destinations, and a plan item retains the folder recorded when it was added. Folder changes affect later additions. File organization settings cannot be changed while tasks are running.

3.3 Extraction and RINEX conversion

The option to ask about extraction and RINEX conversion after downloading is off by default. When enabled, MARK asks once after the observation download plan finishes. Precise products do not use this step. Enable it for ordinary RINEX, or leave it off or cancel the prompt to retain the original compressed files.

(1) Enable the option before starting downloads.

(2) When the observation plan finishes, check the file count, estimated additional space and available disk space in the confirmation. Free space on the indicated drive before retrying if necessary.

(3) After confirmation, MARK extracts files and converts Compact RINEX d / crx to o / rnx. Corresponding compressed inputs and intermediate files are removed only after all outputs have been committed successfully. Unfinished inputs remain after failure or cancellation.

For example, .crx.gz is first extracted to .crx, then converted to .rnx. Extraction alone does not produce ordinary RINEX. Postprocessing covers files downloaded successfully or skipped because they already existed in this task. It does not scan or clean the entire folder. Section 6.6 explains the logic.

4 Observation files

Use the observation workspace in this order: sources → dates → criteria → geographic scope → station selection → download plan → downloading. Start with an indexed date and broad conditions, then add restrictions after confirming that results are available. The overview shows the observation workspace with all 39 locally available sources. BDS daily quality and station selection, and the global versus European geographic-scope example, are separate demonstrations.

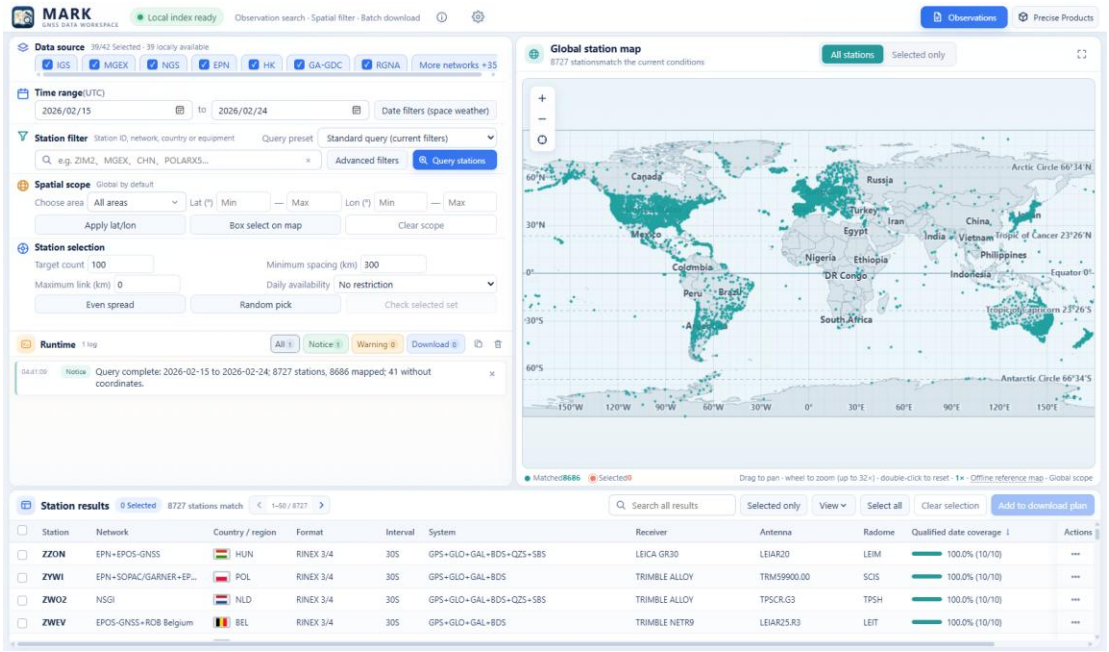


Figure 4.1 Observation workspace with all locally available sources

4.1 Select data sources

(1) Use the common source buttons at the top to select or deselect sources. Open More networks for the complete selector.

(2) Search by source name, choose one or more locally available sources and click Done. Figure 4.2 shows the selection of all locally available sources.

(3) Use Select all networks for all available sources and reverse the selection when the button offers deselection. Check that unindexed sources are not included.

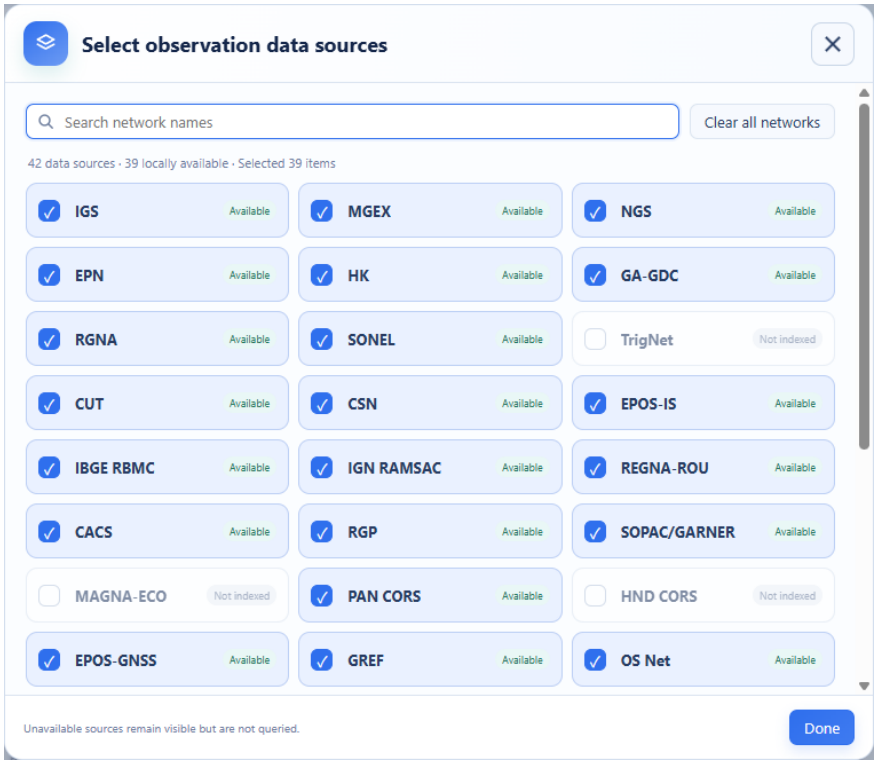


Figure 4.2 Selecting the locally available observation sources

The v1.0.0 interface lists 42 source options, of which 39 have usable local indexes. Available means that an index can be read, rather than that files exist for every date. Selecting an unindexed entry does not build its knowledge base. Chapter 7 lists all sources and download hosts.

4.2 Set UTC dates

Enter start and end dates in Time range (UTC). Both endpoints are included. Set the same date twice for a single-day task. The examples use February 15–24, 2026, covering 10 UTC dates, as shown in Figure 4.3.

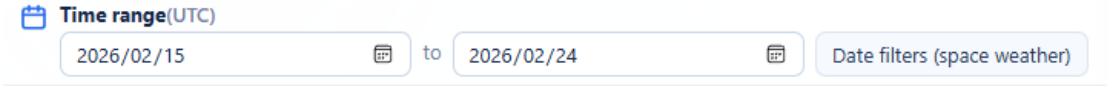


Figure 4.3 UTC start and end dates for observation queries

Type dates directly or use the calendar. Dates are interpreted in UTC, so local midnight is not the UTC day boundary. The knowledge bases cover dates through August 1, 2026, inclusive, but individual sources have different start dates, end dates and gaps. Follow the query messages for actual coverage.

4.3 Advanced date filters

Open Date filters (space weather) to select research dates by space environment. Otherwise leave all three indices unrestricted. Daily maximum Kp, daily minimum Dst and daily F10.7 first determine the dates used by subsequent observation filters.

(1) Choose a keep or exclude rule and enter a threshold or range for each enabled index. A date is retained only if all enabled conditions are met.

(2) Inspect the original, retained, excluded and missing counts, then review excluded dates in the time series.

(3) Use the Kp, Dst and F10.7 buttons to show or hide their curves. Hiding a curve also hides its axis but does not disable the corresponding filter.

(4) For long periods, use zoom controls, the full-period control and date navigation to inspect a shorter interval. Hover over or click a date for values and reasons.

(5) Click Apply and refresh. To remove all date restrictions, clear the conditions and apply again.

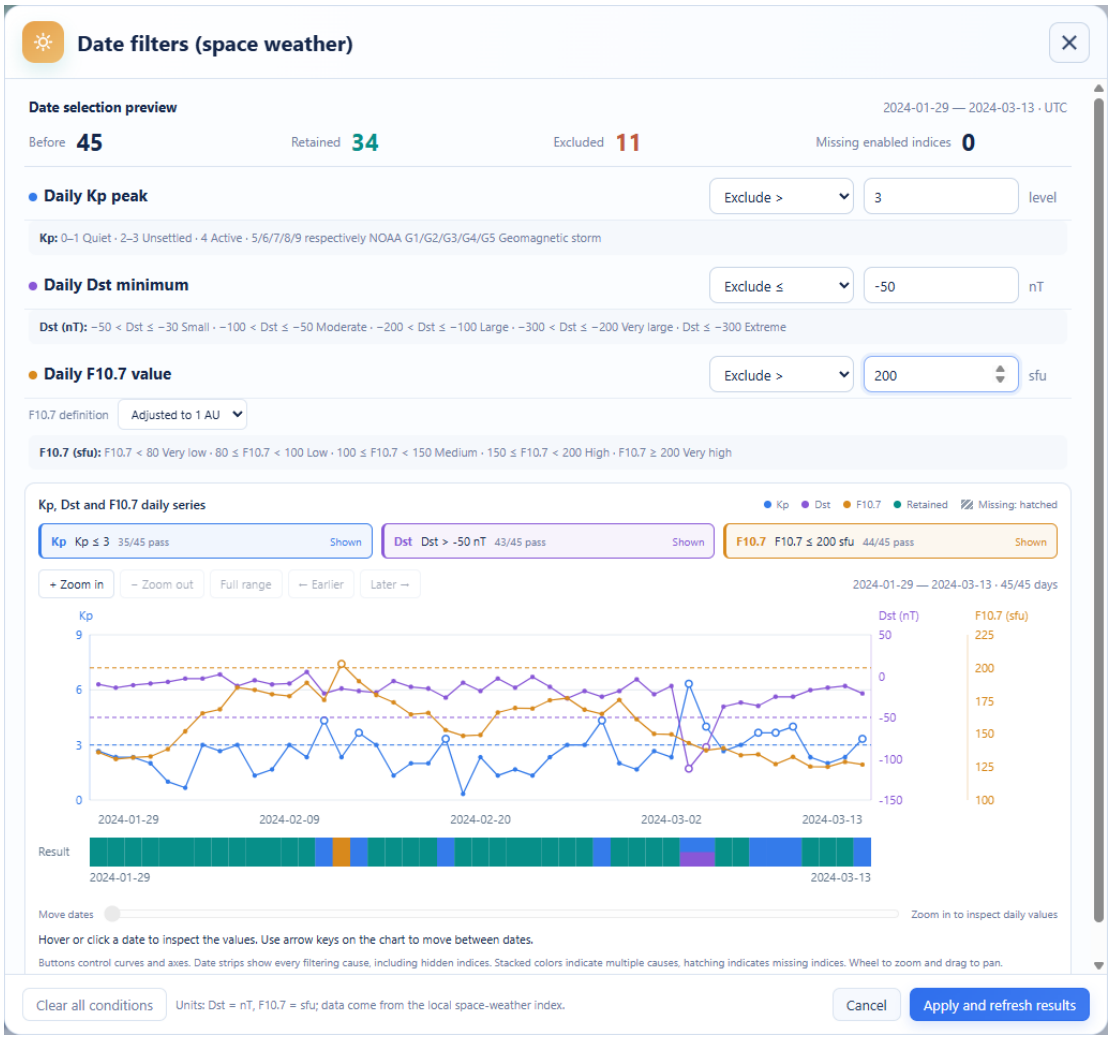


Figure 4.4 Space weather criteria, time series and retained dates

Figure 4.4 is a separate example for January 29–March 13, 2024. Excluding dates with Kp > 3, Dst ≤ −50 nT or F10.7 > 200 sfu retains 34 of 45 dates. These restrictions are not combined with the 10-day BDS example. Green cells indicate retained dates. Other colors identify the excluding indices, dividing a cell equally when several indices exclude that date. Hatching indicates a missing enabled index.

Table 4.1 Space weather conditions

Index	Daily value	Example
Kp	Maximum Kp within the day.	Exclude > 3 retains Kp ≤ 3, including the threshold itself.
Dst	Minimum within the day, in nT.	Exclude ≤ −50 retains Dst > −50 nT.
F10.7	Daily solar radio flux at 10.7 cm, in sfu. Observed and 1 AU adjusted values are available.	Exclude > 200 retains ≤200 sfu. The two flux definitions may retain different dates.

The guide ranges help interpret values and enter thresholds. They do not replace study-specific criteria. To inspect an index without filtering dates by it, leave its rule unrestricted and display its curve.

4.4 Station search and result table

Enter a station ID, network, country code or equipment keyword in Station filter, then click Query stations. For example, ZAMB narrows the search to that station. An empty keyword returns all stations satisfying the other conditions. Keyword, date, advanced and spatial filters act together. The country/region classifications in MARK draw on those used by the IGS and are intended solely for technical interoperability and statistical convenience, without any other implications.

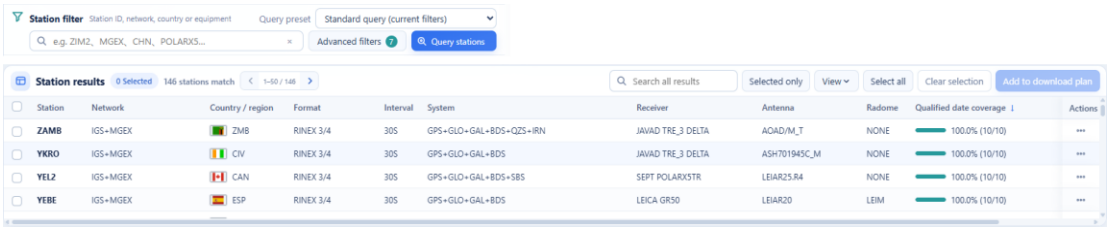


Figure 4.5 Station query controls and the resulting table

Check network, format, interval, constellations, receiver, antenna and qualified date coverage. Scroll or change pages, and use View to configure columns. Search all results narrows the current results, while Selected only shows checked stations. Table and map counts may differ because records without coordinates cannot be plotted.

A row generally summarizes a station over the query period. Equipment and observation attributes may change with date, so a summary or the newest device is not daily evidence for the entire period. Click a row or View station details in its menu to inspect coverage, sources and equipment history.

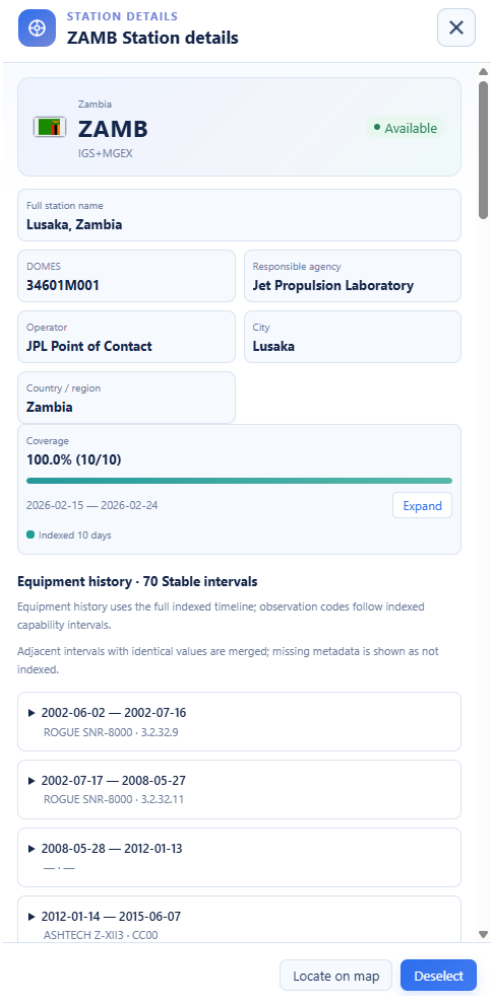


Figure 4.6 Date coverage and equipment history in station details

4.5 Advanced station filters

Open Advanced filters and enable the required criteria. Disabled criteria do not affect eligibility. Figure 4.7 shows availability, format, reference frame and equipment settings; observation codes and daily quality follow below.

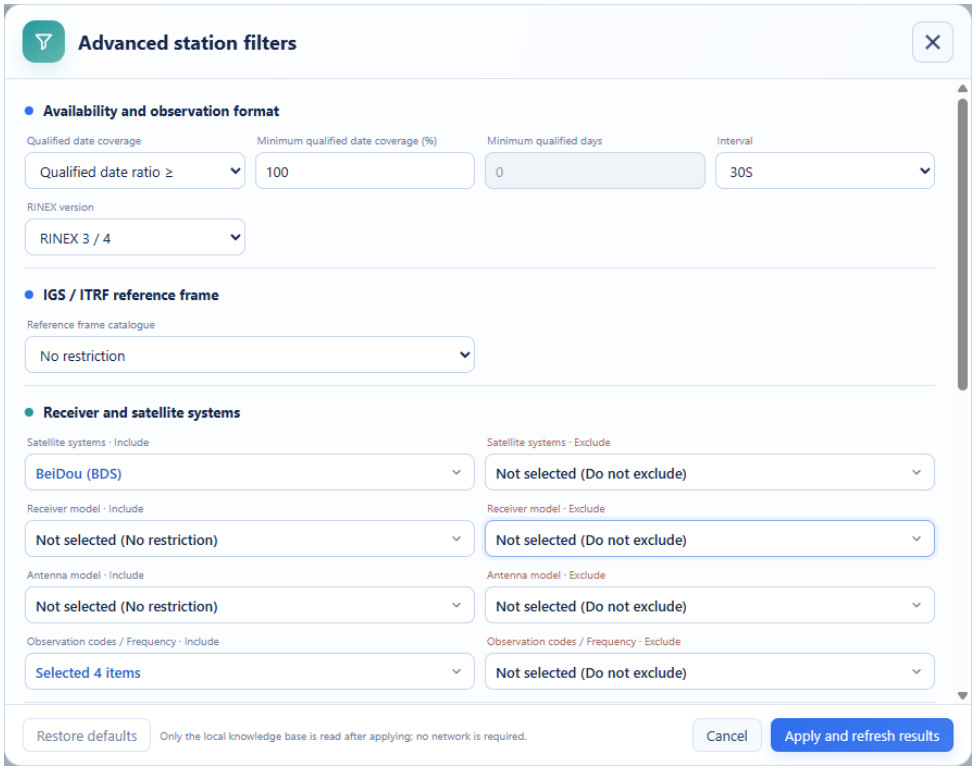


Figure 4.7 Qualified date coverage, format and equipment filters

(1) Under Availability and observation format, choose a qualified date coverage rule and enter a percentage or minimum number of dates, or leave it unrestricted.

(2) Choose an interval, such as 30S for 30 seconds. RINEX options are unrestricted, RINEX 2 and RINEX 3 / 4. Reset restores a 30S interval, rather than clearing every restriction.

(3) Search and select values in the include or exclude lists for constellations, receivers and antennas. Several models can be selected within one category. Section 6.1 explains their combination.

(4) Choose observation codes / frequencies by constellation and RINEX version. Identical code text alone does not establish that signals belong to the same constellation.

(5) Click Apply and refresh. If no initial query has been run, click Query stations in the main workspace to load results.

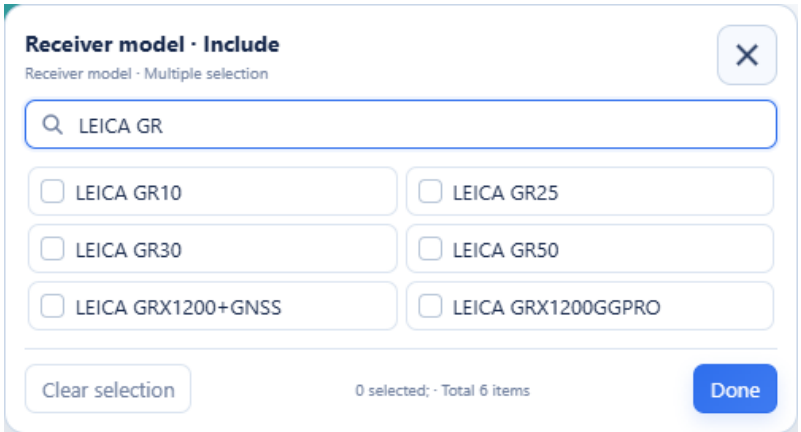


Figure 4.8 Searching related receiver models

Receiver and antenna conditions use records applicable to each date. The IGS/ITRF option selects members of an official core-station list; automatic mode uses IGS14, IGb14, IGS20, IGb20 or IGc20 according to date. It does not transform coordinates. GIM station rules retain or exclude known contributing stations for the selected providers and levels. Section 6.1 explains the union across the query period.

Observation code records and the six quality indicators currently mainly cover IGS/MGEX. Select sources with the required records before enabling these filters. Figure 4.9 explicitly selects BeiDou for completeness and RMS multipath, leaving the other four criteria unrestricted.

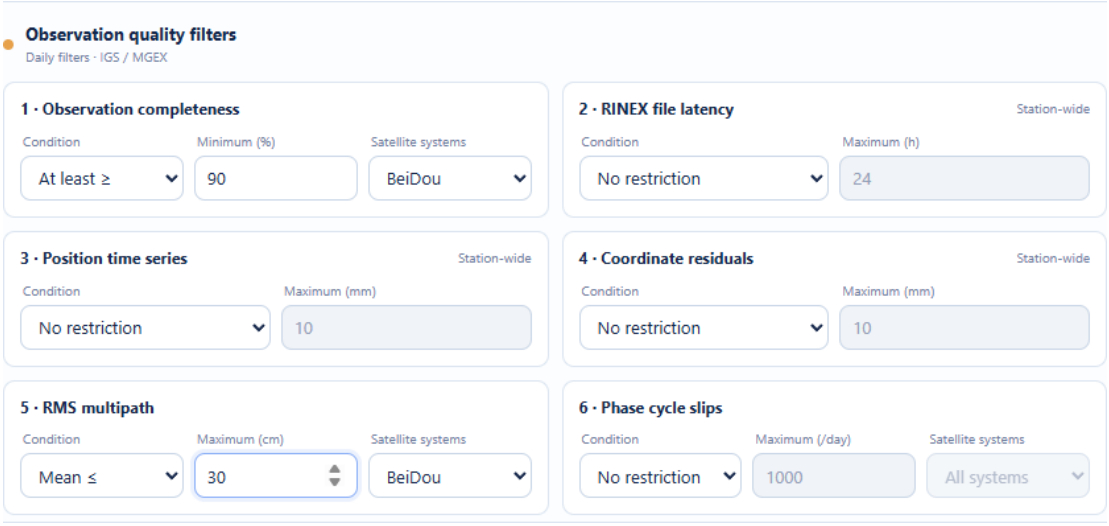


Figure 4.9 Daily quality criteria and independent constellation selectors

Table 4.2 Six daily quality criteria

Criterion	Statistic and unit	Constellation scope
Observation completeness	Actual / expected observations for the date ×100%; must meet the minimum.	All systems, GPS, GLONASS, Galileo, BeiDou or QZSS.
RINEX file latency	Daily file latency, h; must not exceed the threshold.	Station as a whole.
Position time series	Daily mean absolute valid detrended east, north and up components, mm; maximum threshold.	Station as a whole.
Coordinate residuals	Daily mean absolute valid east, north and up residuals, mm; maximum threshold.	Station as a whole.
RMS multipath	Mean RMS across valid frequency bands of the selected system for that date, cm; maximum threshold.	All or one system, independently selected.
Cycle slips	Daily count for the selected system; all systems sums their counts, slips/day; maximum threshold.	All or one system, independently selected.

Every quality criterion is evaluated daily, rather than after averaging across the study period. Completeness, multipath and cycle slips have independent constellation selectors, separate from the constellation include filter. Quality cannot be further separated by observation code. Missing valid records for an enabled indicator make that date unqualified, which indicates insufficient evidence rather than proving poor scientific data quality.

The BDS example uses IGS/MGEX, February 15–24, 2026, 30S, RINEX 3/4, BDS and C2I, L2I, C6I, L6I, BDS daily completeness ≥90%, BDS daily mean RMS multipath ≤30 cm and qualified date coverage of 100%.

This installation returns 146 stations. These thresholds illustrate operation and should be adapted to the intended processing method.

4.6 Geographic scope

Geographic scope narrows the candidate set after observation filtering. This section describes coordinate bounds and a rectangle drawn on the map. Restricting the region is optional. The default scope is global, so users can query without setting a region.

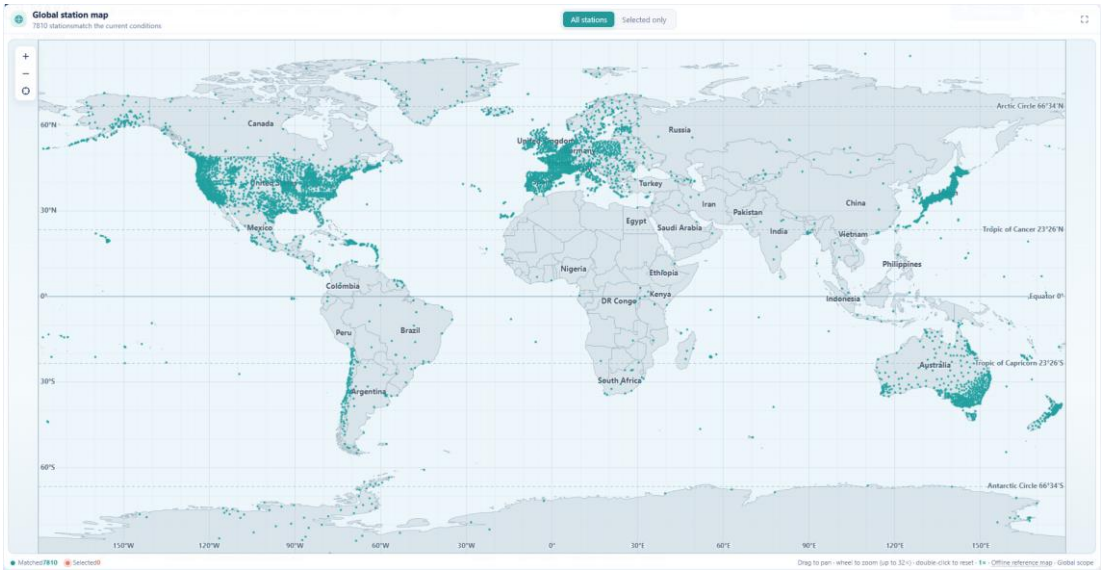
(1) Enter minimum and maximum latitude and longitude and click Apply lat/lon. Latitude 35 to 65 and longitude −15 to 40 define the European example in Figure 4.10.

(2) Click Box select on map and drag a rectangle. Ordinary dragging pans the map, the wheel zooms, and double-clicking resets the view.

(3) Clear scope restores global spatial conditions. Dates, interval and other advanced criteria must be cleared separately.

Figure 4.10 illustrates an independent query for February 15, 2024. Select all 39 locally available sources and keep the default 30 s interval, with other filters unrestricted. The query returns 7,810 stations. Geographic scope is global by default, and restricting the region is optional. To retain only the European example region, enter −15 and 40 as the longitude bounds and 35 and 65 as the latitude bounds, then click Apply lat/lon. The result contains 1,677 stations. Both maps use the same global extent to show the effect of the bounds.

(a) Default global scope: 7,810 stations



(b) 15°W–40°E, 35°N–65°N: 1,677 stations

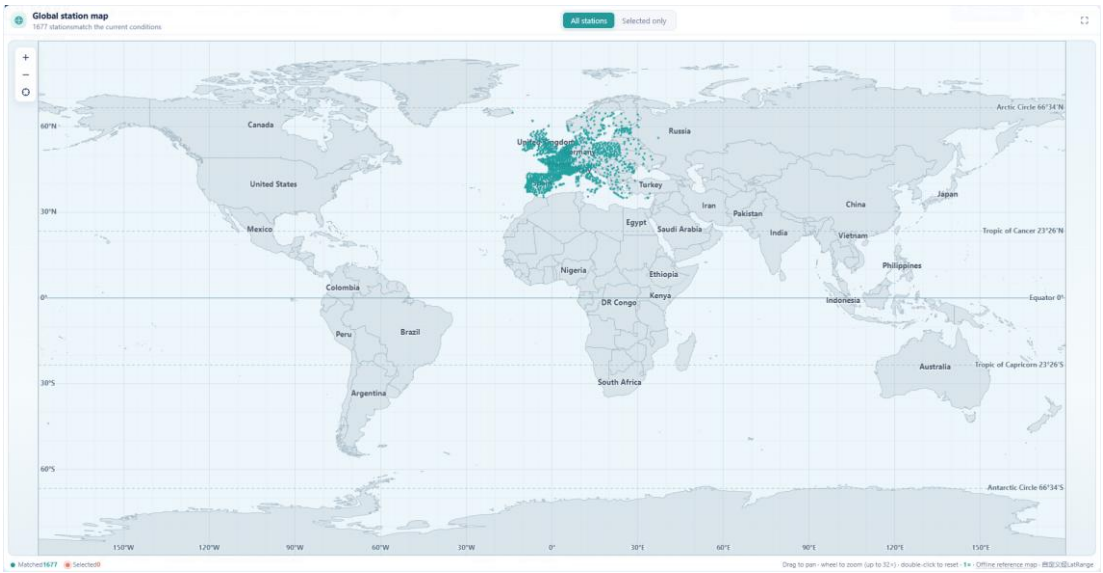


Figure 4.10 Station distributions with global scope and coordinate bounds

Latitude ranges from −90 to 90° and longitude from −180 to 180°. West and south are negative. A western bound greater than the eastern bound, such as 170° to −170°, represents a range across the date line. Check the resulting map against the intended area.

4.7 Select stations

Eligible stations form a candidate set and are not downloaded automatically. Check stations manually or specify a target for even spread or random selection. Figure 4.11 selects 60 of the 146 candidates in the separate BDS example.

(1) Enter Target count. For minimum spacing, enter a distance in kilometres; 0 disables the constraint.

(2) For a connected regional network, set Maximum link. Global selection normally leaves it at 0. Set a minimum daily count or rate under Daily availability if needed.

(3) Click Even spread or Random pick. MARK updates checked stations in the map and table, using different colors for candidates and selected stations.

(4) Click Check selected set to inspect confirmed availability, unknown records, deficient dates and achieved distances. Click dates to inspect their maps or export the report.

(5) If count, distance and daily conditions cannot all be met, inspect the report. A target count alone does not establish that the selected set is valid.

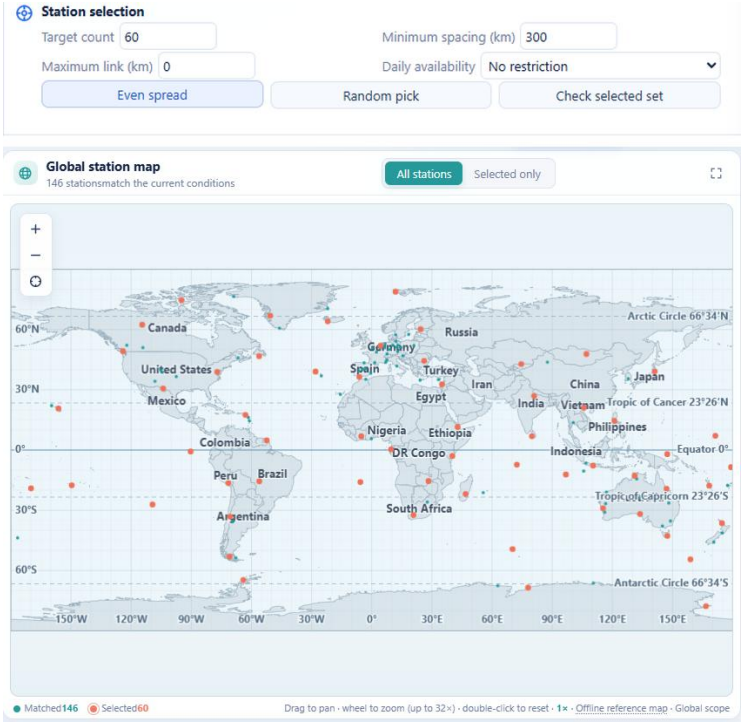


Figure 4.11 Even spread settings and the selected stations

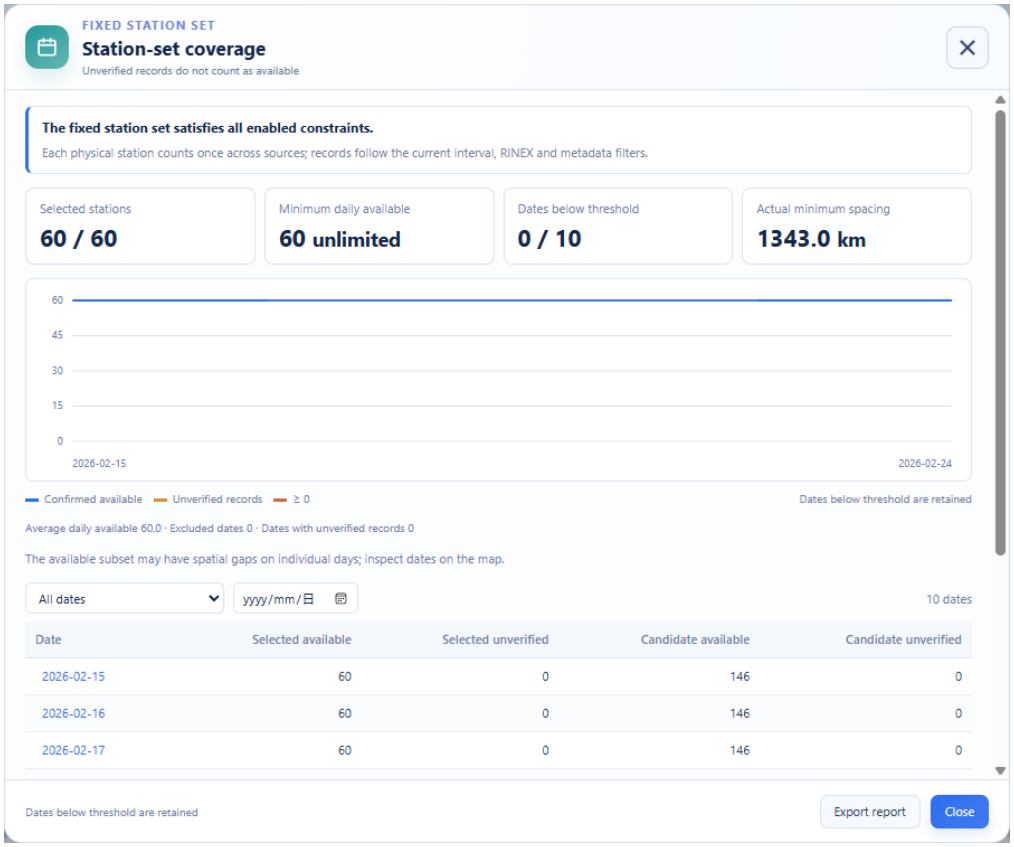


Figure 4.12 Daily availability check for a fixed station set

Qualified date coverage evaluates one station across dates. Daily availability evaluates how many selected stations are usable on each date. These constraints are not interchangeable. MARK does not discard deficient dates to make a selection pass. Section 6.2 explains the selection method.

4.8 Add a download plan

(1) Check the intended stations in the table. Use Browse... at the bottom to choose a writable destination.

(2) Click Add to download plan to record stations, query dates and filtering criteria in the observation plan. This step does not transfer files.

(3) Open the cart at the bottom, inspect station IDs, dates and pending states, and check the items to execute, as shown in Figure 4.13.

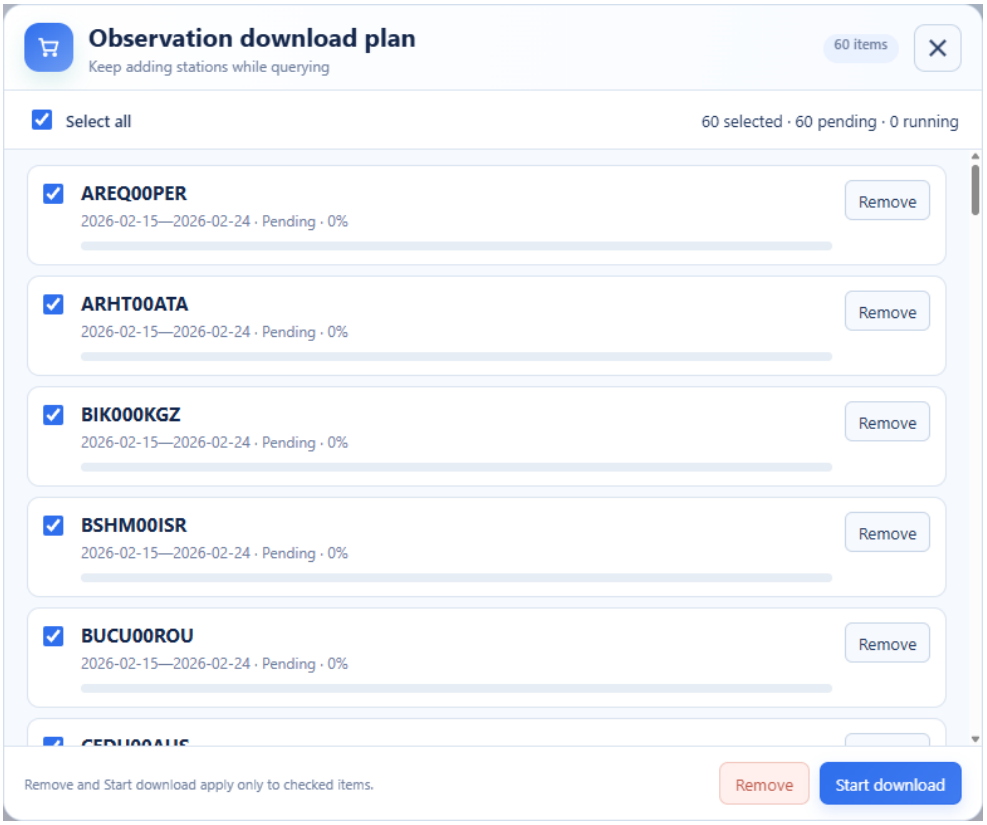


Figure 4.13 Stations, dates and states in the observation plan

Select all covers all items in the current workspace's plan, including items not currently visible. Start download and Remove act on checked items. Removing an item does not delete saved files. Plan items retain their original dates and criteria; changing a later query does not edit an existing plan.

4.9 Download and inspect results

Confirm the destination and plan, then click Start download in the plan or bottom bar. MARK resolves actual files before transferring them. The bottom bar reports progress, the plan shows item states, and Runtime records messages. Figure 4.14 shows the controls.

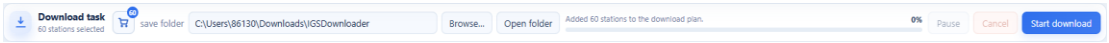


Figure 4.14 Destination, plan, progress and download controls

(1) Continue querying, adjusting criteria and adding tasks during transfers. Observation and product plans are managed separately.

(2) Pause and Resume control the current workspace. Cancel ends its current batch while retaining completed files. Plan checkboxes do not stop items already transferring.

(3) Inspect successful, existing, failed and cancelled counts. Check the destination and logs subfolder. A progress value of 100% does not mean every file succeeded.

(4) If postprocessing was enabled, follow the extraction and RINEX conversion prompt. Check space, retain error information and inspect output headers.

Restarting MARK does not restore closed plans or running tasks. Query and add tasks again; existing files are checked before skipping. Pause / Resume does not guarantee byte-range resumption on every server.

4.10 Observation presets

Query presets configure several conditions together for repeated preparation. Select a preset, set UTC dates and query. Presets may change sources, advanced conditions and selection parameters. A preset is not a scientific method by itself; check and adapt its thresholds.

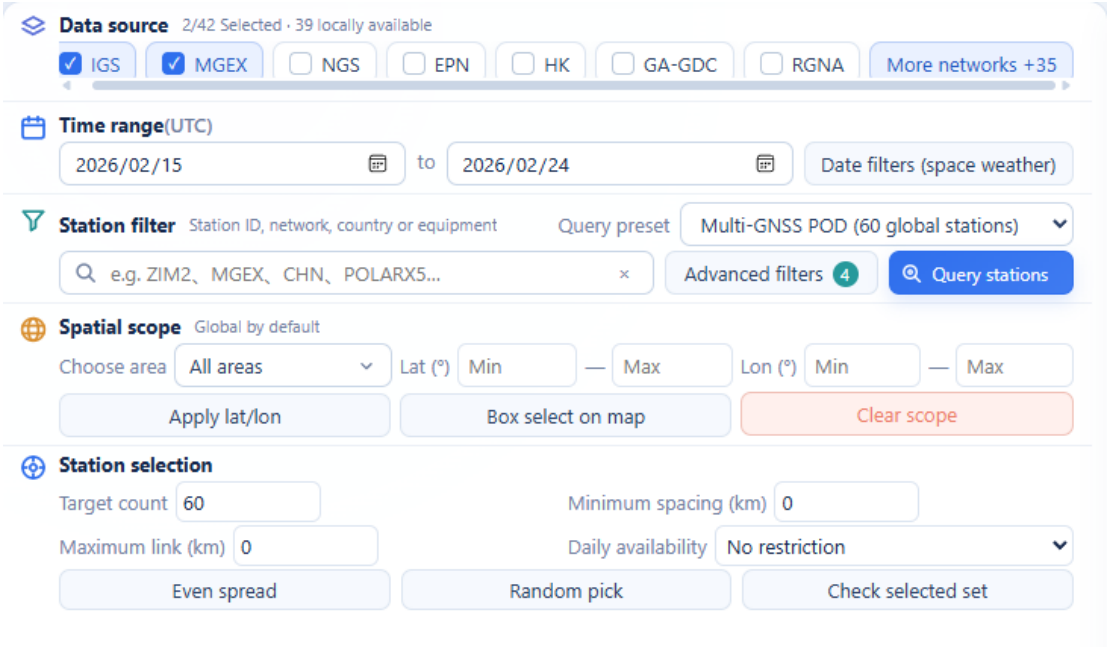


Figure 4.15 Observation preset and the updated controls

Table 4.3 Observation presets

Preset	Main conditions	Selection and notes
Standard query	Retains current sources, dates, equipment, quality and scope.	Does not restore all defaults. Clear relevant conditions separately for an unrestricted query.
IGS/ITRF reference frame	IGS, 30S, RINEX3/4; official core-station list appropriate to the date.	Restores global scope, leaves qualified date coverage unrestricted by default, and does not fix a station count.
GIM ionosphere stations	IGS/MGEX, GPS, 30S; retains known contributors used by the chosen providers during the period.	Target 250 with automatic even spread. Its internal 245-station daily query condition differs from the main selection field; inspect controls and reports.
GIM evaluation	All locally available sources, GPS, 30S, RINEX3/4, coverage ≥90%; excludes selected GIM contributors.	50 stations, 50 km minimum spacing and at least 35 confirmed stations daily; automatic even spread. Independence depends on contributor record coverage.
Multi-GNSS POD	IGS/MGEX, GPS+GLONASS+Galileo+BDS, 30S, RINEX3/4, completeness ≥95%.	Target 60 with automatic even spread. Different from the manually configured BDS example in Section 4.5. Inspect quality constellation selectors.

Preset	Main conditions	Selection and notes
Network RTK candidates	All locally available sources, GPS+GLONASS, 30S, RINEX3/4.	Retains user scope and target; maximum link 75 km. Select manually after querying. Does not perform RTK processing or stream corrections.

Check which fields changed before querying. GIM station, GIM evaluation and POD presets can select an even spread automatically after querying. Standard, reference frame and Network RTK presets require manual selection or a selection button. Some quality constellation choices and values may persist between presets. Inspect actual settings rather than relying on a preset name. No preset starts downloading automatically; inspect the selection and add a plan.

5 Precise products

Switch to Precise Products at the top right. Set dates, filter by category, level and analysis center, check availability, then select products and add a download plan. The final sections describe quick presets for repeated tasks.

5.1 Set UTC dates

Enter start and end dates in Time range (UTC), including both endpoints. Use one date for a single-day task. Observation and product calendars are separate, so check dates after switching workspace. The product example covers January 1–February 1, 2024, comprising 32 UTC dates.

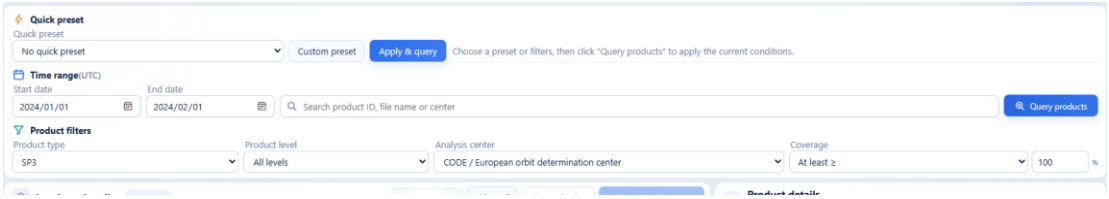


Figure 5.1 Product dates, category, level, center and coverage

5.2 Filter categories levels and analysis centers

(1) Select SP3, CLK or another Product type, or all products. The interface has 13 categories, described in Section 7.2.

(2) Select Final, Rapid, Ultra-rapid or another Product level. Level describes production or release, rather than download speed.

(3) Select the producing institution or service under Analysis center. Use the search box for product IDs or filename keywords when targeting a product family.

(4) Click Query products. Figure 5.2 shows an ordinary query for CODE SP3 products.

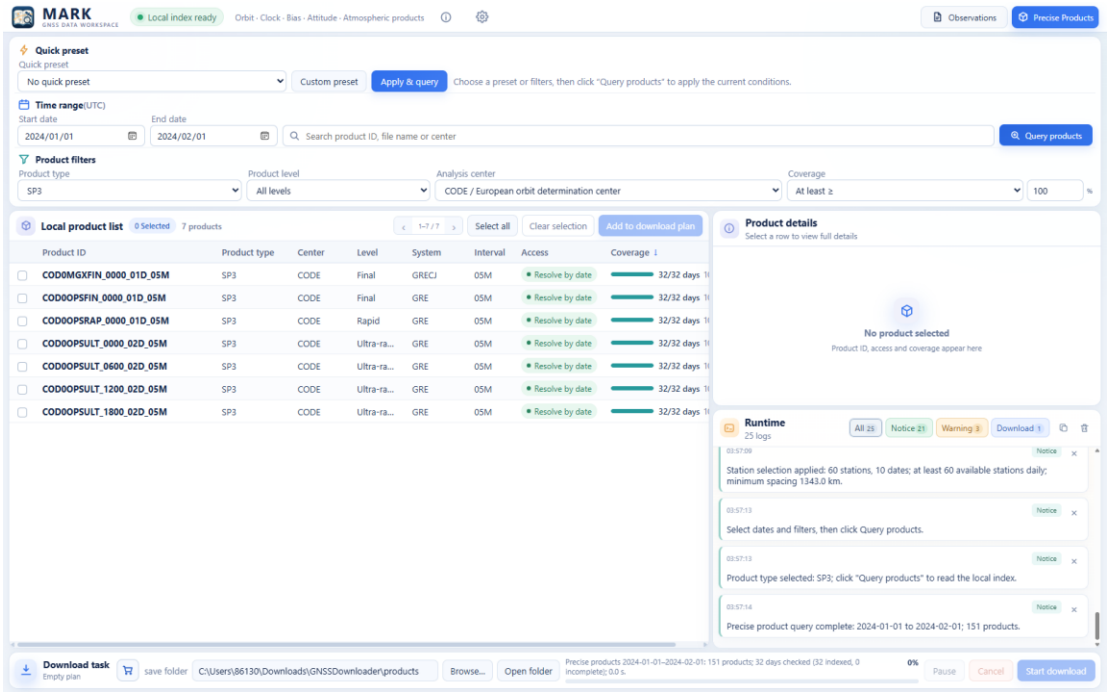


Figure 5.2 Ordinary query results for CODE SP3 products

An analysis center produces a product, while a download source supplies the file. They may differ. Main filters have no separate constellation or sampling interval dropdowns. Check those attributes in results or details; use a custom preset when reusing interval rules.

5.3 Restrict availability

Coverage can be unrestricted, a minimum percentage or a minimum number of available dates. Choose a mode, enter the value and query again. The 100% example in Figure 5.1 requires recorded coverage for every requested date.

A result such as 32/32 days, 100% describes record coverage across the query dates. It does not establish epoch completeness, product accuracy or compatibility between products. Static files and files spanning multiple dates use their own availability rules; a new file is not necessarily required each day.

5.4 Inspect products and add a plan

A row generally represents a product series expanded over the requested dates, rather than a downloaded individual file. Click a row for center, level, constellation, interval, date coverage and sources, as shown in Figure 5.3. Placeholder dates or 0000 in a series ID do not replace query dates; actual filenames are resolved when the plan executes.

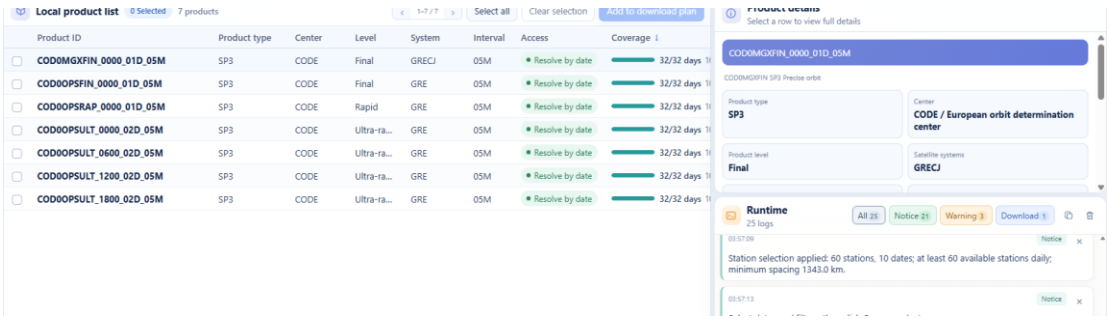


Figure 5.3 Product attributes, coverage and details

(1) Check product family, constellations and interval. For joint SP3, CLK and bias use, follow downstream requirements for signals, reference frame and bias conventions.

(2) Check the required products, choose a product destination and click Add to download plan.

(3) Open the cart, inspect product IDs, dates and pending items, check the items and click Start download, as shown in Figure 5.4.

(4) Inspect states and logs as in Section 4.9. The product plan is independent of the observation plan and does not run observation extraction or RINEX conversion.

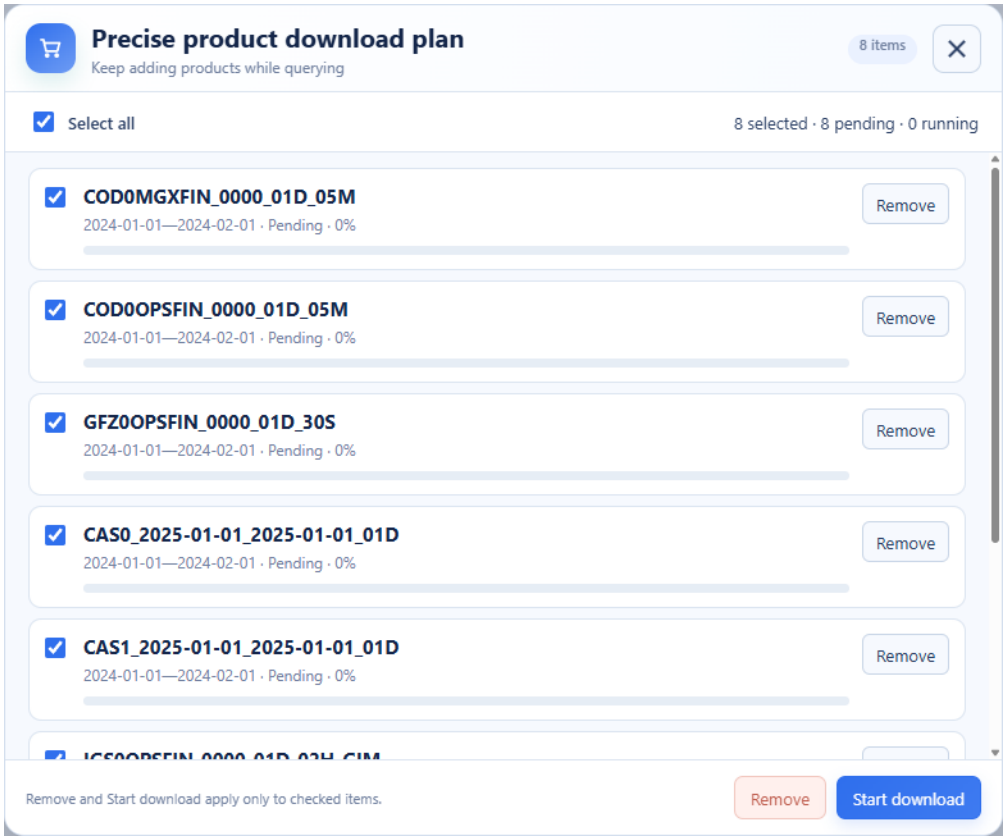


Figure 5.4 Precise product download plan

Antenna model declarations in CLK details help check ANTEX. When only a base model is declared and a historical revision is unconfirmed, MARK reports that uncertainty. A current ATX file with the same name must not be assumed identical to the historical processing model.

5.5 Built-in quick presets

After setting dates, select a Quick preset and click Apply preset and query. Changing the dropdown alone does not generate a product set. Ordinary Query products uses main filters and is not equivalent to applying a preset. Figure 5.5 uses the IGS final PPP preset.

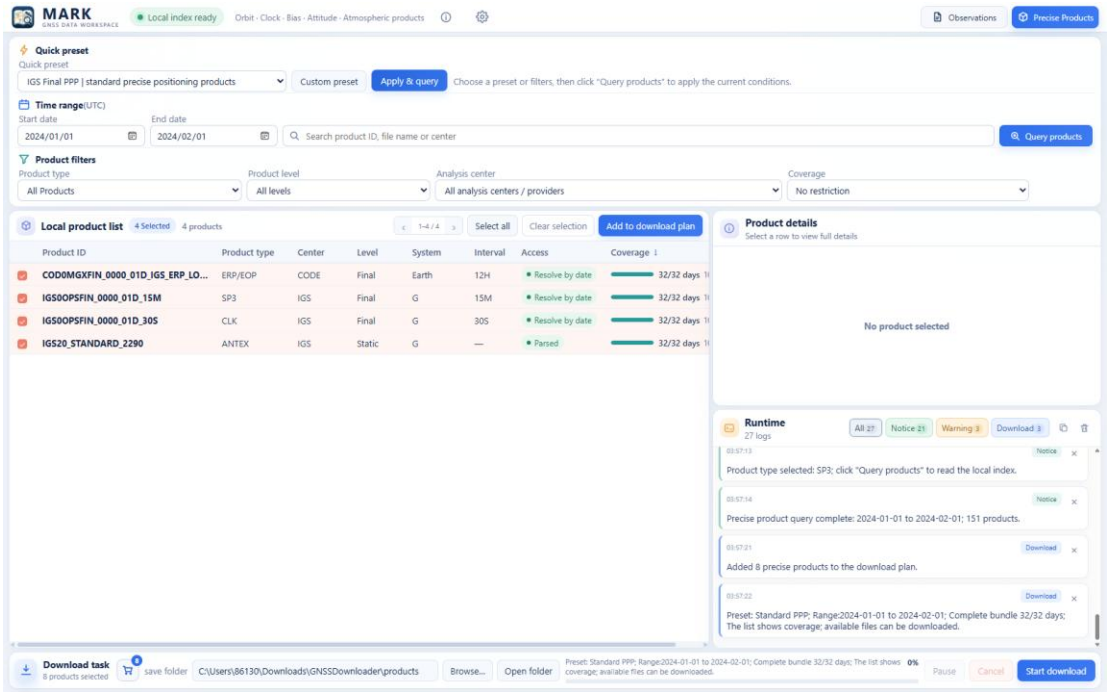


Figure 5.5 Built-in quick preset and returned product series

Table 5.1 Built-in product presets

Preset	Products and selection principles
PRIDE PPP-AR	WUM0MGXRAP Rapid multi-GNSS orbit, 30S clock, matching OSB, ERP and ORBEX. Matches ANTEX to daily CLK declarations and checks SINEX, GIM, VMF and BRDC auxiliaries.
IGS final PPP	IGS Final OPS orbit, 30S clock, matching ERP/EOP and IGS GLOBAL ANTEX for standard float PPP preparation.
UPD PPP-AR	SP3, CLK, UPD, ERP and ANTEX from matching providers and product families. Some UPD records describe local inventory without online download routes.
OSB PPP-AR	SP3, 30S CLK, OSB, ERP and ANTEX from the same provider and product line. Compares full-set coverage among identifiable Final/Rapid candidates.
PPP-RTK	Prepares basic SP3, 30S CLK, OSB, ERP and ANTEX files. Regional atmospheric corrections, SSR streams and network processing must be obtained separately.
Generic PPP smart selection	Compares SP3, CLK, ERP and ANTEX, prioritizing full-set date coverage before compatibility evidence, level and CLK interval. Rapid may precede Final when its coverage is better.

A complete set requires all members of that preset on the same date. Per-row coverage describes only one product series. PRIDE also checks two SINEX types and four epochs each of VMF1 and VMF3; a missing auxiliary epoch can leave the set incomplete. Inspect Runtime gaps, compatibility messages and member lists, rather than orbit and clock availability alone.

PRIDE matches ANTEX using the selected daily CLK model declarations, preferring explicit filenames or historical revisions such as IGS20_2290. It does not simply select the current igs20.atx by observation date. Check

unresolved model evidence before formal processing. Quick presets organize files; they do not perform PPP, ambiguity fixing or a PPP-RTK service.

5.6 Custom quick presets

Use Custom presets to save multiple product rules for repeated queries. Rules may use different centers and their matches are merged. Figure 5.6 illustrates five rules.

(1) Open Custom presets, choose New preset and enter an identifiable name.

(2) Set type, level, center and applicable interval for each rule. Add product adds a rule; interval controls are disabled for inapplicable categories.

(3) Save the preset. For later use, set dates, choose its name from Quick preset and click Apply preset and query.

(4) Inspect every match, select the required series and add a plan. Reopen Custom presets to edit rules or manage saved presets.

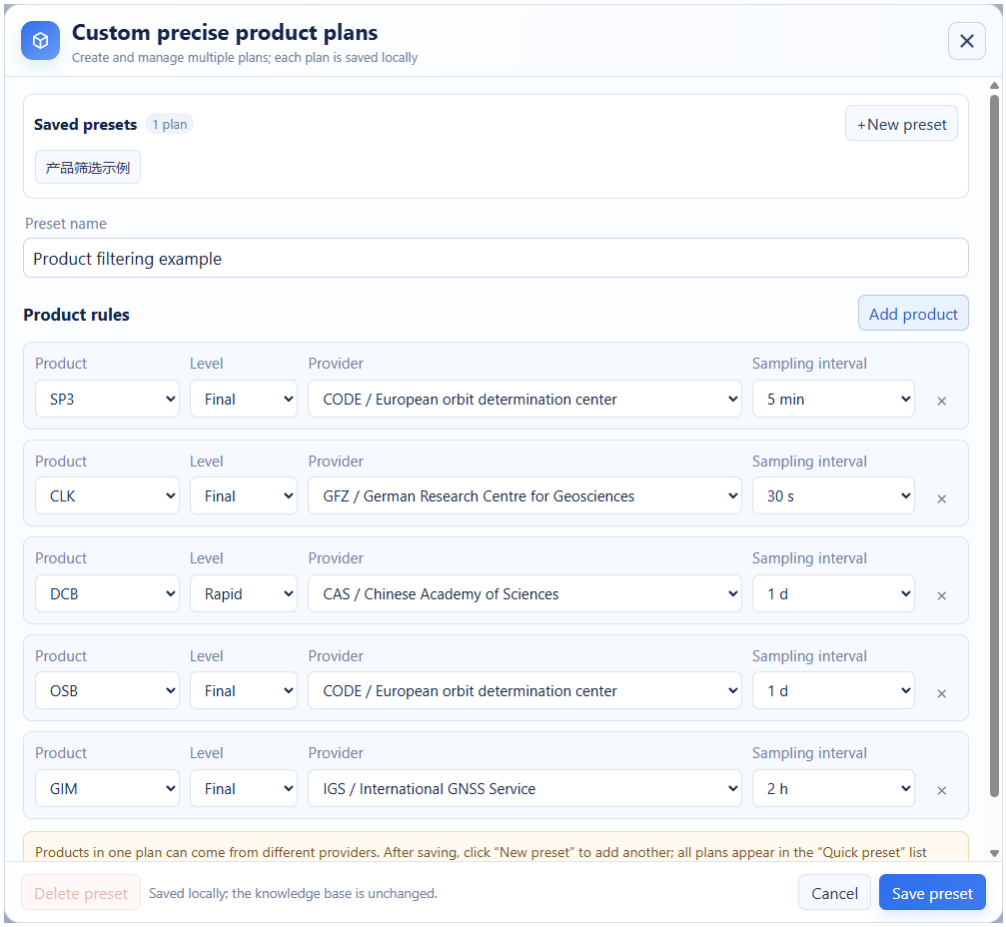


Figure 5.6 Custom rules for five product categories

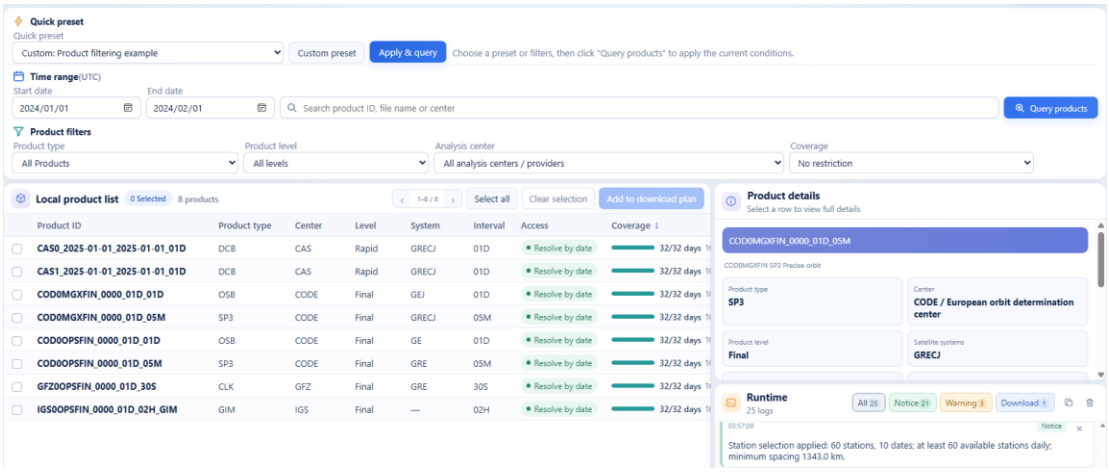


Figure 5.7 Actual series matched by the custom rules

One rule can match several series, so five rules need not return five rows. Custom presets reuse rules without ensuring compatibility across centers. Figure 5.6 is a query example, not a recommendation to mix CODE orbits and GFZ clocks. Presets are saved in local WebView2 user data without changing the knowledge bases and may need recreating after moving computers or clearing that data.

6 Detailed logic

6.1 Daily filtering and qualified date coverage

The basic record unit is a station and a UTC date. MARK first selects sources and a date range, then applies space weather or other date rules to obtain the effective date set. For each station and effective date, it matches files, interval, RINEX version and applicable equipment and observation attributes. Enabled quality criteria are intersected separately for that date.

Different criterion categories must be satisfied together. Multiple included constellations must all be present. Multiple included receivers or antennas within one category are alternatives. Exclusion rules remove matches. Observation codes carry constellation and RINEX version meaning and require corresponding historical evidence; a matching code name from another constellation cannot substitute.

Qualified date coverage is the percentage of effective query dates on which a station satisfies all enabled criteria. A date without a file record, with failing files or with missing required quality evidence is unqualified. Missing disabled indicators do not affect eligibility.

Let T be the effective query date set. Set $a_{s,t}$ to 1 when station s has file records satisfying all enabled criteria on date t, and 0 otherwise. Qualified date coverage C_s, in percent, is:

$$C_s = \frac{100}{|T|}\sum_{t \in T} a_{s,t} \tag{6.1}$$

Table 6.1 Date coverage and observation indicators

Indicator	Numerator and denominator	Object evaluated
Qualified date coverage	Qualified dates / effective query dates.	One station across the study period.
Daily observation completeness	Actual / expected observations for that date.	Observation count for one station and selected constellation within a day.

Indicator	Numerator and denominator	Object evaluated
Daily availability constraint	Confirmed selected stations on a date, or their proportion of the selected set.	The usable size of a fixed station set on each date.

For example, 27 qualified dates out of 30 effective dates give 90% coverage. Dates removed by space weather filters leave the denominator. Effective dates with missing files, failed conditions or missing required quality records remain in it. A 100% requirement means that every effective date qualifies, not that every observation epoch is complete.

Quality indicators are taken from upstream public records and are not recalculated from newly downloaded files. All systems completeness sums actual and expected counts across systems. RMS multipath averages valid bands within the selected system and date; upstream zero multipath values are unavailable, while zero can be valid for other indicators. Time series and residual values average valid directional components within a day, never across dates.

Reference frame membership follows list validity dates. GIM contributor rules use the union of known station identifiers for the selected providers and levels over the whole query period, rather than matching every observation date separately. Contributor counts without station identifiers cannot exclude particular stations. Unknown equipment or indicators cannot satisfy strict inclusion rules automatically, nor do they prove the original observations unusable.

6.2 Even spread and distance constraints

Even spread selects a spatially dispersed subset of the current candidates. MARK computes spherical distances from latitude and longitude and uses farthest point sampling to avoid repeatedly choosing nearby stations in dense regions. Candidate distribution and coordinate evidence limit the result. The method does not guarantee equal-area coverage or a globally optimal set for every scientific objective.

Let C denote the candidates, with latitude φ_i and longitude λ_i in radians. MARK first converts each position to a unit vector:

$$u_i = (\cos\varphi_i \cos\lambda_i, \cos\varphi_i \sin\lambda_i, \sin\varphi_i) \tag{6.2}$$

Spherical distance is calculated below, using R = 6371.0088 km and clamping to limit numerical error. Squared chord distance q_{ij} gives the same ordering, avoiding inverse trigonometric evaluation for every comparison.

$$q_{ij} = \mathrm{clamp}(2 - 2u_i \cdot u_j, 0, 4), \quad d_{ij} = 2R \arcsin(\frac{\sqrt{q_{ij}}}{2}) \tag{6.3}$$

With no maximum-link constraint and a target of at least two stations, the basic spatial pass starts from the farthest candidate pair:

$$(i_0, j_0) \in \underset{i,j \in C, i \neq j}{\arg\max} d_{ij} \tag{6.4}$$

Let S_k be the selected set and F_k the unselected candidates satisfying the current distance constraints. Each step chooses the candidate whose nearest selected station is farthest away, until the target is reached or no feasible candidate remains:

$$x_{k+1} \in \underset{x \in F_k}{\arg\max} \min_{s \in S_k} d(x, s) \tag{6.5}$$

With maximum link enabled, MARK first restricts candidates to the largest connected component, starts near its centroid and adds stations connected to the selected set within the threshold. The formulas describe the basic

spatial pass. Daily constraints add coverage-aware candidate construction and replacement repair; the final report determines whether the set passes.

Minimum spacing limits the distance between every pair of selected stations. Maximum link is a connectivity constraint for regional networks: stations within the threshold are connected by edges, and selection uses a connected candidate region and checks the selected network. It does not require every pair of stations to lie within the maximum link. A small link threshold is generally unsuitable for global coverage. Zero disables either constraint.

For final set S and user threshold d_{min}, minimum spacing is expressed below. Maximum link instead requires connectivity of the threshold graph; it does not bound the largest distance between every pair.

$$\min_{i,j \in S, i \neq j} d_{ij} \geq d_{min} \tag{6.6}$$

With daily availability enabled, MARK searches and checks a fixed station set using the filtered station-date records. Unknown records do not count as confirmed availability. Deficient dates are not dropped to meet the threshold. Records without coordinates may appear in the table, but cannot contribute invented locations to distance optimization.

Daily availability A_t sums confirmed qualified station-date records and must meet minimum count M on every effective date. For a rate p%, MARK uses target station count K and rounds p×K/100 up to obtain M. Unknown records do not contribute.

$$A_t = \sum_{s \in S} a_{s,t} \geq M, \ \forall t \in T \tag{6.7}$$

Random selection also respects requested count, spatial limits and enabled constraints. Repeated runs may differ. Retain station IDs, query settings and exported reports for reproducibility. Recheck after manually changing stations or criteria because an earlier report may be outdated.

6.3 Plans and actual file lists

Checking results, adding plans, resolving file lists and transferring files are separate steps. Each plan item records station or product ID, dates, destination and conditions. At execution, the observation module reads qualified station-date records and rechecks interval, format and historical attributes. The product module resolves actual files by series and date. These records supply filenames, download addresses and known alternatives.

A physical station may occur in several sources but should not be counted repeatedly in the selected set. MARK resolves logical files rather than downloading duplicates merely to retain more source labels. Alternative routes depend on actual stored records; servers must not be assumed interchangeable.

The workspaces maintain separate plans, states and destinations. Queries remain independent of transfers, allowing further queries and additions during downloading. A plan's destination is fixed when added; switching workspace or changing the current folder does not move existing files.

6.4 Parallel transfers and adaptive scheduling

Several plan items and several files within an item can run concurrently. Both workspaces share a global limit of 12 simultaneous transfers, subject to lower per-host limits. The number of active plan items is therefore not the number of active transfers, and a file is not split into 12 segments.

Let H be the active hosts, c_h the number of files transferring from host h and L_h its current adaptive limit. Scheduling respects both global and host limits:

$$\sum_{h \in H} c_h \leq 12, \quad 0 \leq c_h \leq L_h \tag{6.8}$$

Concurrency is managed separately for each target host. MARK raises it gradually when transfers remain stable and throughput improves. It lowers concurrency, and may cool down, when rate limits, temporary connection errors or diminishing throughput appear. Files from different sources are interleaved so one slow host does not occupy every transfer slot.

Protocols and services have different initial and maximum concurrency. SFTP and some limited servers are conservative, whereas public object storage can allow more transfers. The program manages these settings automatically. Settings has no manual thread-count field, and increasing the target station count does not raise the global cap.

6.5 Retries alternatives and task control

Ordinary libcurl transfers retry recoverable errors a limited number of times, usually at most three attempts including the first, with short waits between attempts. Timeouts, interrupted connections, temporary server errors and rate limiting can be recoverable. Specialized source adapters can use different strategies, so three attempts is not a universal rule.

Explicit missing-file or permission responses, such as HTTP 404, 401, 403 or FTP 550, normally lead to another recorded route rather than repeated waiting on the same one. A file fails when no usable route remains. Historical records do not ensure that a server still retains or permits access to the file.

Received data are written to temporary files, checked for basic content after transfer and committed to final names. Task state is checked while paused, cancelled or waiting for transfer capacity. Cancellation retains completed files, but does not guarantee a partially downloaded file suitable for byte-range resumption. Resumption depends on the server and actual file state.

Inspect per-file outcomes, routes and error summaries in logs under the plan destination. Diagnose the file and server from those records before repeatedly clicking Start download.

6.6 Existing files and postprocessing

MARK performs basic content and format checks before skipping existing files. Converted ordinary RINEX may satisfy a corresponding compressed observation task. Basic checks detect some empty files, error pages and malformed content, but do not replace scientific validation of epochs, signals, orbits or biases.

Observation postprocessing supports GZIP, Unix compress (.Z), ZIP, TAR/TAR.GZ, BZIP2, XZ and nested compression. Bundled CRX2RNX converts Compact RINEX. Each input is handled in an isolated temporary location. Inputs are cleaned only after all outputs are committed, and existing outputs with different content are not overwritten.

Postprocessing uses limited CPU-dependent parallelism, up to four input files at once. Extraction, conversion and validation proceed sequentially within each input, followed by name-conflict checks during commit. Insufficient space pauses processing and allows retry or cancellation. Space estimates are conservative, rather than exact final sizes.

6.7 Troubleshooting order

Table 6.2 Troubleshooting

Symptom	Check first
Blank window or missing component	Windows x64, WebView2 Runtime, complete software/resources and accompanying libraries.
No query results	Use an indexed source and covered date; clear keywords, date, equipment, quality and spatial conditions, then restore them individually.
Many stations lost after quality filtering	Selected constellations, daily indicator coverage and thresholds. Missing records alone do not prove poor quality.
Fewer map points than table rows	Missing or unknown coordinates. Do not infer locations from station IDs.
Cannot reach target count	Candidate count, minimum spacing, maximum link and daily constraints; inspect the selection report.
404 FTP 550 or permission errors	Actual filenames, dates, routes and access requirements. Historical indexing does not prove current download access.
Files still have crx or YYd suffixes	Whether postprocessing was enabled and completed. Extraction alone is not ordinary RINEX conversion.
Progress finished with failures	Individual plan states and log summaries. Rerun failed items and inspect actual files.

7 Data sources

7.1 Observation sources and download hosts

Table 7.1 lists all 42 observation source options in this v1.0.0 installation. IGS/MGEX can be networks or data views, while SOPAC/GARNER and EPOS-GNSS can aggregate several networks. The option count is not a count of independent physical networks. Thirty-nine options are locally available; TrigNet, MAGNA-ECO and HND CORS are unindexed.

Hosts identify current download adapters or archive entry points; users do not need to construct paths manually. Some European sources discover original files through EPOS GLASS, with downloads served by member institutions. Knowledge base records and logs determine the actual file route.

Table 7.1 Observation source options and download hosts

UI name	Coverage or institution	Host	Local state
IGS	International GNSS Service; global reference stations	CDDIS, IGN, BKG, WHU, GFZ, ESA and others	Available
MGEX	IGS Multi-GNSS Experiment and multi-system records	Recorded archive routes shared with IGS	Available
NGS	US NOAA National Geodetic Survey CORS	noaa-cors-pds.s3.amazonaws.com	Available
EPN	EUREF Permanent GNSS Network	igs.bkg.bund.de	Available

UI name	Coverage or institution	Host	Local state
HK	Hong Kong SatRef reference station service	rinex.geodetic.gov.hk	Available
GA-GDC	Geoscience Australia data center	ga-gnss-data-rinex-v1.s3.amazonaws.com	Available
RGNA	Mexico INEGI active geodetic network	geodesia2.inegi.org.mx (SFTP)	Available
SONEL	GNSS records supporting sea-level studies	ftp.sonel.org	Available
TrigNet	South African continuous reference network	ftp.trignet.co.za	Unindexed
CUT	Curtin University GNSS records	saegnss02p.curtin.edu.au	Available
CSN	University of Chile national seismic GNSS network	gps.csn.uchile.cl	Available
EPOS-IS	Iceland EPOS and meteorological records	data.epos-iceland.is; api.vedur.is	Available
IBGE RBMC	Brazil IBGE continuous monitoring network	geoftp.ibge.gov.br	Available
IGN RAMSAC	Argentina IGN continuous satellite monitoring	ramsac.ign.gob.ar	Available
REGNA-ROU	Uruguay IGM active geodetic network	sftp.igm.gub.uy; pp.igm.gub.uy	Available
CACS	Canadian Active Control System; NRCan	cacsa.nrcan.gc.ca	Available
RGP	France IGN permanent GNSS network	rgpdata.ensg.eu; rgp.ign.fr	Available
SOPAC/ GARNER	Scripps Orbit and Permanent Array Center archive	garner.ucsd.edu	Available
MAGNA-ECO	Colombia IGAC continuous GNSS network	srvstorage.igac.gov.co	Unindexed
PAN CORS	Panama ANATI continuous reference stations	cors.anati.gob.pa (SFTP)	Available
HND CORS	Honduras continuous reference station records	www.ign.hn	Unindexed
EPOS-GNSS	European EPOS-GNSS Data Gateway	gnssdata-epos.oca.eu and member file servers	Available
GREF	BKG German integrated geodetic reference network	igs.bkg.bund.de	Available
OS Net	UK Ordnance Survey GNSS reference network	EPOS GLASS member download routes	Available

UI name	Coverage or institution	Host	Local state
ERGNSS	Spain IGN continuous GNSS network	EPOS GLASS member download routes	Available
RENEP	Portugal DGT permanent GNSS network	EPOS GLASS member download routes	Available
GeoNet NZ	New Zealand GeoNet and PositioNZ records	geonet-open-data.s3.ap-southeast-2.amazonaws.com	Available
RING	Italy INGV national integrated GNSS network	gnss.isprambiente.it and recorded routes	Available
APOS	Austria BEV positioning service	gnss.bev.gv.at and EPOS member routes	Available
ROB Belgium	Royal Observatory of Belgium GNSS records	gnss.be and EPOS member routes	Available
DPGA	Delft Permanent GNSS Array, Netherlands	DPGA records at gnss1.tudelft.nl	Available
CatNet	Catalonia ICGC GNSS network	geofons.icgc.cat	Available
SPSLux	Luxembourg ACT satellite positioning service	data.public.lu; download.data.public.lu	Available
KASI-NGII	Korea KASI archive and NGII station portal	gnss-ftp.kasi.re.kr; actual KASI file list	Available
RENAG	French permanent GNSS research network	renag.resif.fr	Available
NSGI	Netherlands geodetic infrastructure collaboration	gnss-data.kadaster.nl	Available
NOA GNSS	National Observatory of Athens GNSS network	www.gein.noa.gr	Available
REDGAE	Spanish regional GNSS network aggregation	Public EPOS GLASS member routes	Available
GEONET JP	Geospatial Information Authority of Japan GEONET	terras.gsi.go.jp (anonymous HTTPS workflow)	Available
DLAB	Delft University of Technology GNSS field laboratory	gnss1.tudelft.nl	Available
MAGNET	Nevada Geodetic Laboratory regional archive	geodesy.unr.edu	Available
PANGA/CWU	Pacific Northwest Geodetic Array; CWU	panga.org; geodesy.org and recorded mirrors	Available

IGS/MGEX records can provide several archive routes, including CDDIS, WHU, IGN, BKG, GFZ, ESA, KASI and SOPAC/GARNER. Coverage differs by date, RINEX version and high-rate data. No one host should be treated as a universal mirror.

7.2 Precise and auxiliary products

The interface groups products into 13 categories. Table 7.2 distinguishes content from common file representations. Analysis centers include producing institutions and services, rather than exclusively observation data centers.

Table 7.2 Product categories and file representations

Category	Content	Common representation
SP3	Precise satellite orbits	SP3
CLK	Precise satellite clocks	RINEX clock / CLK
ERP/EOP	Earth rotation and orientation	ERP and EOP series
DCB	Differential code biases	DCB, Bias-SINEX
OSB	Observable-specific biases	BIA, Bias-SINEX
UPD	Uncalibrated phase delays	WL, NL and EWL products
GIM	Global ionospheric maps	IONEX, commonly INX/IOX suffixes
TRO	Tropospheric parameters	TRO-SINEX, ZPD
SINEX	Coordinates and solution products	SNX, SSC
ANTEX	Antenna phase center calibration	ATX
ORBEX	Satellite attitude	ORBEX, commonly OBX
BRDC	Broadcast ephemerides	RINEX navigation
VMF	Tropospheric mapping grids	VMF1 and VMF3 epoch grids

GIM describes content and IONEX a common exchange format; these are not separate categories. ORBEX is the exchange representation for attitude files. Check actual series and file coverage rather than assuming that category names imply support for every satellite or signal.

Table 7.3 Product institution and service options

Name or code	Interface description
IGS	International GNSS Service
CODE	European orbit determination center
EMR/NRCan	Natural Resources Canada
ESA	European Space Agency
GFZ	German Research Centre for Geosciences
GRG	France GRGS
JPL	Jet Propulsion Laboratory
MIT	Massachusetts Institute of Technology
NGS	U.S. National Geodetic Survey
SIO	Scripps Institution of Oceanography
USNO	U.S. Naval Observatory
WHU/WUM	Wuhan University

Name or code	Interface description
JGX	Japanese multi-GNSS joint analysis
HUST	Huazhong University of Science and Technology
IERS	International Earth Rotation and Reference Systems Service
IAC	Russian Information and Analysis Center
JAXA	Japan Aerospace Exploration Agency
SHAO	Shanghai Astronomical Observatory
BKG	German Federal Agency for Cartography and Geodesy
IGN	French National Geographic Institute
DLR	German Aerospace Center
CAS	Chinese Academy of Sciences
UPC	Polytechnic University of Catalonia
OPTIMAP	OPTIMAP Center
Other	Other providers
TU Wien	Vienna University of Technology
WCC	Wuhan Combination Center
ASI	Italian Space Agency
AUS	Geoscience Australia
BEK	Bavarian Academy of Sciences, Geodesy and Glaciology
BEV	Austrian Federal Office of Metrology and Surveying
CSR	Center for Space Research, University of Texas at Austin
DEO	Delft Institute for Earth-Oriented Space Research
EUR	EUREF / EPN combined solutions
GOP	Geodetic Observatory Pecny
GSI	Geospatial Information Authority of Japan
IGE	Instituto Geografico Nacional, Spain
LPT	Federal Office of Topography swisstopo
MUT	Military University of Technology, Poland
NCL	Newcastle University
NKG	Nordic Geodetic Commission
OLG	Institute for Space Research, Graz
PDR	GFZ / TU Dresden joint reprocessing
PGC	Pacific Geoscience Centre, Canada
RGA	Republic Geodetic Authority, Serbia
ROB	Royal Observatory of Belgium
SGO	Satellite Geodetic Observatory, Hungary

Name or code	Interface description
SUT	Slovak University of Technology
TUG	Graz University of Technology
TUM	Technical University of Munich
ULR	University of La Rochelle / IGN consortium
UPA	University of Padova
WUT	Warsaw University of Technology
GIA	Geophysical Institute, University of Alaska Fairbanks
MCC	Russian Mission Control Center
EPN	EUREF Permanent GNSS Network

Table 7.4 Main product archives and distribution routes

Entry	Description
CDDIS: cddis.nasa.gov / gdc.cddis.eosdis.nasa.gov	Historical IGS and other providers' orbits, clocks and biases. Applicable authentication and protocol requirements remain in force.
IGN: igs.ign.fr	IGS and other institutions' archives; paths come from file records.
BKG: igs.bkg.bund.de	IGS and EUREF material and selected product distribution.
WHU: igs.gnsswhu.cn; public PRIDE routes	Wuhan University and other indexed routes. A WHU label does not make product families and services interchangeable.
CODE GFZ and other official archives	Some products come directly from producers; others use central archives or mirrors.
VMF: vmf.geo.tuwien.ac.at	VMF models and epoch grids. Check model, grid and solution type.

Some entries describe local inventory without an online address, particularly UPD. Their appearance in results does not guarantee that MARK can download them. Original providers determine access, licensing and attribution.

7.3 Historical attributes quality and date records

Table 7.5 Sources of filtering information

Information	Source and use
Equipment and observation attributes	Equipment installation/removal dates from Site Logs, supplemented by RINEX headers where historical coverage is missing. Applicable dated records support equipment and constellation filters.
Observation codes and frequencies	Extracted from indexed IGS/MGEX RINEX headers. Constellation and RINEX version jointly define a code.
Six daily quality indicators	Public IGS Network charts, for example https://network.igs.org/ABMF00GLP . Includes observation counts, latency, position time series, residuals, multipath and cycle slips.

Information	Source and use
Reference frame station lists	Bundled IGS14, IGb14, IGS20, IGb20 and IGc20 core-station lists and validity dates.
GIM contributors	Station identifiers in recorded GIM-related information, limited by provider, level and date coverage.
Kp Dst and F10.7	Daily historical space weather index. Kp uses daily maxima, Dst daily minima, and F10.7 retains observed and 1 AU adjusted values for the chosen definition.
Offline reference map	Bundled map and source notices for location and spatial filtering reference.

IGS chart identifiers are rnxobs, latency, timeseries, residuals, rms-multipath and cycle-slips. Use them to check upstream statistical definitions. Upstream records can change, while local knowledge bases are release snapshots. Downloading a new file does not recompute or fill its historical quality indicators.

7.4 Coverage and access requirements

This release includes 64 DuckDB knowledge bases with an overall cutoff of August 1, 2026, inclusive. Individual sources can start later, end earlier or have gaps. Querying does not update every knowledge base automatically. Use matching packages supplied by the maintainer for new software or data releases.

Local index availability and remote download access are separate states. HTTPS, FTP, FTPS and SFTP have different access requirements. Japanese GEONET uses the supported anonymous Terras HTTPS workflow; SFTP requires the Windows OpenSSH client and provider permission. MARK Settings has no dedicated account or proxy configuration page. Follow provider rules for restricted sources.

For reproducibility, retain software and knowledge base versions, UTC dates, sources, criteria, selected station or product IDs, actual filenames and relevant hashes. Include errors and logs in issue reports after removing passwords, tokens and other private information.